

Prefactor Coefficient Laws for the Gauss–Legendre AGM Algorithm

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Abstract

We study two fitted coefficients associated with the Gauss–Legendre arithmetic–geometric mean (AGM) algorithm for π . For consecutive Gauss–Legendre approximants π_n , define

$$\alpha_n = -\frac{\pi_{n+1} - \pi}{\pi_{n+1} - \pi_n}.$$

This coefficient is the exact one-step weight which corrects π_{n+1} to π using the increment $\pi_{n+1} - \pi_n$.

The first result is a coefficient law for the Gauss–Legendre error operator:

$$\frac{\alpha_{n+1}}{\alpha_n^2} \longrightarrow \frac{1}{2}.$$

The proof uses Brent’s theta–function expansion for the lower Gauss–Legendre approximant. In the notation of the present paper, that expansion yields the asymptotic

$$\pi - \pi_n = (2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O\left(2^n e^{-\pi 2^{n+2}}\right).$$

Consequently, if

$$\varepsilon_n = \pi_n - \pi = A_n e^{-\pi 2^{n+1}},$$

then

$$A_n \sim -2^{n+4}\pi^2,$$

and hence $A_{n+1}/A_n \rightarrow 2$. The limiting coefficient $1/2$ thus arises from exact cancellation of the doubly–exponential factor in the quotient α_{n+1}/α_n^2 , leaving a prefactor ratio that tends to $1/2$.

For the companion numerator–channel coefficients β_n , a direct theta–function analysis gives the sharper prefactor law

$$\beta_n = -\left(2^{n+2}\pi - 1\right) e^{-\pi 2^{n+1}} \left(1 + O\left(e^{-\pi 2^{n+1}}\right)\right),$$

and consequently

$$-\pi 2^{n+1} \frac{\beta_{n+1}}{\beta_n^2} \longrightarrow 1.$$

Thus the previously observed decay $\beta_{n+1}/\beta_n^2 \rightarrow 0$ is analytic, with an explicit leading scale. Finally, we explain why the static Baker N–series cancellation operator does not transfer directly to the Gauss–Legendre setting, while the natural dynamic cancellation principle is closely related to Aitken’s Δ^2 process.

1 Preliminaries

Let $\{\pi_n\}$ denote the Gauss–Legendre π –sequence generated by the AGM iteration

$$\begin{aligned} a_{n+1} &= \frac{1}{2}(a_n + b_n), & b_{n+1} &= \sqrt{a_n b_n}, \\ t_{n+1} &= t_n - p_n(a_n - a_{n+1})^2, & p_{n+1} &= 2p_n, \end{aligned}$$

with initial values $a_0 = 1$, $b_0 = 1/\sqrt{2}$, $t_0 = 1/4$, $p_0 = 1$. The Gauss–Legendre approximant is

$$\pi_n = \frac{(a_n + b_n)^2}{4t_n} = \frac{a_{n+1}^2}{t_n}.$$

Write $\varepsilon_n = \pi_n - \pi$ for the signed error. For the lower Gauss–Legendre approximant considered here one has $\pi_n < \pi$ and hence $\varepsilon_n < 0$.

The Gauss–Legendre algorithm is classically known to converge extremely fast (the number of correct digits roughly doubles per iteration). A more precise description of the error is doubly exponential. Brent’s theta–function analysis gives a sharp expansion for the lower Gauss–Legendre approximant; see Brent [4, Section 4, Eq. (20), pp. 9–10]:

$$\pi - \pi_n = (2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}}).$$

Remark 1. The phrase “quadratic convergence” should not be confused with a fixed constant- C model for every associated error sequence. For Brent’s upper approximants a_n^2/t_n , the corresponding errors $e_n = a_n^2/t_n - \pi$ satisfy

$$\frac{e_{n+1}}{e_n^2} \longrightarrow \frac{1}{8\pi};$$

see Brent [4, Section 4, following Eq. (19)]. By contrast, for the lower approximants used here, $\varepsilon_n = \pi_n - \pi$ satisfies $\varepsilon_{n+1}/\varepsilon_n^2 \rightarrow 0$. Thus the lower channel has a rapidly varying prefactor even though the overall iteration remains quadratically convergent in the usual digits-doubling sense.

2 Fitted Operator Coefficients

Definition 1. Define

$$\alpha_n = -\frac{\pi_{n+1} - \pi}{\pi_{n+1} - \pi_n} = -\frac{\varepsilon_{n+1}}{\varepsilon_{n+1} - \varepsilon_n},$$

so that the exact one-step identity

$$\pi = \pi_{n+1} + \alpha_n(\pi_{n+1} - \pi_n)$$

holds for each n .

Thus α_n is the exact fitted coefficient which recovers π from the two successive approximants.

For later comparison, define the numerator sequence $Q_n = (a_n + b_n)^2$ and the numerator-channel coefficient

$$\beta_n = -\frac{Q_{n+1} - 4\pi t_{n+1}}{Q_{n+1} - Q_n},$$

which plays an analogous role on the numerator channel.

3 Reduction to Error Ratios

Lemma 1. Assume $r_n := \varepsilon_{n+1}/\varepsilon_n \rightarrow 0$. Then

$$\alpha_n \sim \frac{\varepsilon_{n+1}}{\varepsilon_n} = r_n,$$

and

$$\frac{\alpha_{n+1}}{\alpha_n^2} \sim \frac{r_{n+1}}{r_n^2} = \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3}.$$

Proof. From $\alpha_n = \frac{r_n}{1-r_n}$ and $r_n \rightarrow 0$ we obtain $\alpha_n \sim r_n$. More precisely,

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{r_{n+1}}{r_n^2} \frac{(1-r_n)^2}{1-r_{n+1}}.$$

Since $r_n, r_{n+1} \rightarrow 0$, the correction factor tends to 1, and therefore

$$\frac{\alpha_{n+1}}{\alpha_n^2} \sim \frac{r_{n+1}}{r_n^2} = \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3}.$$

□

4 Prefactor Asymptotic

Lemma 2. For the lower Gauss–Legendre approximants one has

$$\varepsilon_n = \pi_n - \pi = -(2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}}).$$

Consequently, writing $\varepsilon_n = A_n e^{-\pi 2^{n+1}}$, we have

$$A_n = -(2^{n+4}\pi^2 - 8\pi) + O(2^n e^{-\pi 2^{n+1}}),$$

so $A_n \sim -2^{n+4}\pi^2$ and hence $A_{n+1}/A_n \rightarrow 2$.

Proof. Since

$$a_{n+1} = \frac{a_n + b_n}{2},$$

the approximant used in this paper satisfies

$$\pi_n = \frac{(a_n + b_n)^2}{4t_n} = \frac{a_{n+1}^2}{t_n}.$$

Brent writes the same Gauss–Legendre auxiliary sequence as s_n rather than t_n , so the present approximant is Brent’s lower Gauss–Legendre approximant

$$\frac{a_{n+1}^2}{s_n}.$$

Brent’s theta-function expansion for this lower approximant gives

$$\pi - \frac{a_{n+1}^2}{s_n} = (2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}});$$

see Brent [4, Section 4, Eq. (20), pp. 9–10]. Therefore

$$\pi - \pi_n = (2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}}).$$

Since

$$\varepsilon_n = \pi_n - \pi,$$

we obtain

$$\varepsilon_n = - (2^{n+4}\pi^2 - 8\pi) e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}}).$$

Writing

$$\varepsilon_n = A_n e^{-\pi 2^{n+1}},$$

gives

$$A_n = - (2^{n+4}\pi^2 - 8\pi) + O(2^n e^{-\pi 2^{n+1}}).$$

Hence

$$A_n \sim -2^{n+4}\pi^2,$$

and therefore

$$\frac{A_{n+1}}{A_n} \rightarrow 2.$$

□

5 The α_n Coefficient Law

Theorem 1. *With α_n as above, one has*

$$\lim_{n \rightarrow \infty} \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2}.$$

Proof. By Lemma 2 we may write $\varepsilon_n = A_n e^{-\pi 2^{n+1}}$ with $A_{n+1}/A_n \rightarrow 2$. Hence $\varepsilon_{n+1}/\varepsilon_n \rightarrow 0$, so Lemma 1 applies and

$$\frac{\alpha_{n+1}}{\alpha_n^2} \sim \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3}.$$

Inserting the asymptotic form cancels the exponential contributions exactly (the exponent combination equals zero), leaving

$$\frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3} \sim \frac{A_{n+2}A_n^2}{A_{n+1}^3} = \frac{A_{n+2}}{A_{n+1}} \left(\frac{A_n}{A_{n+1}} \right)^2.$$

Using $A_{n+1}/A_n \rightarrow 2$ we get

$$\frac{A_{n+2}}{A_{n+1}} \rightarrow 2, \quad \frac{A_n}{A_{n+1}} \rightarrow \frac{1}{2},$$

and therefore the prefactor quotient tends to $2 \cdot (\frac{1}{2})^2 = \frac{1}{2}$. The theorem follows. □

Remark 2. The limit $1/2$ is a prefactor fingerprint of the AGM structure: it comes from exact cancellation of the doubly-exponential factor together with the linear-in- 2^n growth of A_n . It does not follow from a fixed quadratic recurrence $\varepsilon_{n+1} \sim C\varepsilon_n^2$.

6 Numerical Verification of the α_n Law

All reported numerical values were computed with `mpmath` at 500 decimal digits, reconstructing the Gauss–Legendre sequence from its defining recursion and using `mp.pi` for the reference value of π . The fitted coefficients and diagnostic ratios support the theoretical limit over the precision and ranges shown in Tables 1 and 2. As an independent check, the direct error quotient $\varepsilon_{n+2}\varepsilon_n^2/\varepsilon_{n+1}^3$ approaches α_{n+1}/α_n^2 rapidly. The two quantities are not identical at finite n ; their exact relation is

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3} \frac{(1-r_n)^2}{1-r_{n+1}}, \quad r_n = \frac{\varepsilon_{n+1}}{\varepsilon_n}.$$

The correction factor tends to 1, explaining the increasing numerical agreement in Table 2.

n	α_n
0	$4.476840443051657 \times 10^{-3}$
1	$7.278747106485182 \times 10^{-6}$
2	$2.482705775153228 \times 10^{-11}$
3	$2.988090953871379 \times 10^{-22}$
4	$4.397065311352923 \times 10^{-44}$
5	$9.594611535813435 \times 10^{-88}$
6	$4.585616315217315 \times 10^{-175}$

Table 1: Fitted coefficients α_n .

n	α_{n+1}/α_n^2	A_{n+1}/A_n
0	0.3631728700311416	2.3866261558728965
1	0.4686104763344650	2.0871750987545307
2	0.4847784478028806	2.0414374856886031
3	0.4924649101466708	2.0202981874959604
4	0.4962511920458477	2.0100471245391349
5	0.4981302572056423	2.0049984522335209

Table 2: Coefficient and prefactor ratios: the left column approaches 1/2 from below; the right column approaches 2, consistent with the prefactor asymptotic.

7 The Numerator-Channel Coefficients

For this section let Brent’s elliptic nome be

$$q = e^{-\pi},$$

and define

$$x_n = q^{2^{n+1}} = e^{-\pi 2^{n+1}}, \quad C_n = 2^{n+2}\pi - 1.$$

Also set

$$M = \text{AGM}\left(1, \frac{1}{\sqrt{2}}\right) = \theta_3^{-2}(q).$$

The capital notation $Q_n = (a_n + b_n)^2$ distinguishes the numerator sequence from the nome q .

Lemma 3. As $n \rightarrow \infty$,

$$Q_{n+1} - Q_n = -32M^2x_n(1 + 2x_n + 4x_n^2 + O(x_n^4)),$$

and

$$Q_{n+1} - 4\pi t_{n+1} = -32M^2C_nx_n^2 \left[1 + \left(2 - \frac{1}{C_n} \right) x_n^2 + O(x_n^4) \right].$$

The implied constants may be chosen independently of n .

Proof. Brent's theta parametrization gives

$$a_n = \frac{\theta_3^2(q^{2^n})}{\theta_3^2(q)} = M\theta_3^2(q^{2^n});$$

see Brent [4, Section 4, Eq. (13)]. Since $a_{n+1} = (a_n + b_n)/2$,

$$Q_n = 4a_{n+1}^2 = 4M^2\theta_3^4(x_n).$$

Because $x_{n+1} = x_n^2$,

$$Q_{n+1} - Q_n = 4M^2 [\theta_3^4(x_n^2) - \theta_3^4(x_n)].$$

From the defining theta series,

$$\theta_3(x) = 1 + 2x + 2x^4 + O(x^9),$$

hence

$$\theta_3^4(x) = 1 + 8x + 24x^2 + 32x^3 + 24x^4 + 48x^5 + O(x^6).$$

It follows that

$$\theta_3^4(x^2) - \theta_3^4(x) = -8x - 16x^2 - 32x^3 + O(x^5),$$

which proves the first expansion.

For the second difference, Brent's exact limit and tail formulas are

$$t_\infty = \frac{M^2}{\pi}, \quad t_{n+1} - t_\infty = M^2 \sum_{m=n+1}^{\infty} 2^m \theta_2^4(q^{2^{m+1}});$$

see Brent [4, Section 4, Eqs. (17)–(18)]. Therefore

$$Q_{n+1} - 4\pi t_{n+1} = 4M^2 \left[\theta_3^4(x_n^2) - 1 - \pi \sum_{m=n+1}^{\infty} 2^m \theta_2^4(q^{2^{m+1}}) \right].$$

The defining series gives

$$\theta_2(y) = 2y^{1/4} (1 + y^2 + O(y^6)), \quad \theta_2^4(y) = 16y + 64y^3 + O(y^5).$$

Reindexing the tail with $m = n + 1 + j$ yields

$$\sum_{m=n+1}^{\infty} 2^m \theta_2^4(q^{2^{m+1}}) = 2^{n+5}x_n^2 + 2^{n+6}x_n^4 + O(2^n x_n^6),$$

where the terms $j \geq 2$ are smaller still. Also

$$\theta_3^4(x_n^2) - 1 = 8x_n^2 + 24x_n^4 + O(x_n^6).$$

Thus

$$\begin{aligned}\frac{Q_{n+1} - 4\pi t_{n+1}}{4M^2} &= (8 - 2^{n+5}\pi)x_n^2 + (24 - 2^{n+6}\pi)x_n^4 + O(2^n x_n^6) \\ &= -8C_n x_n^2 - 8(2C_n - 1)x_n^4 + O(2^n x_n^6).\end{aligned}$$

Since $C_n \asymp 2^n$, division by the leading term gives the stated second expansion. $\square$

Theorem 2 (Numerator-channel prefactor law). *The numerator-channel coefficient satisfies*

$$\boxed{\beta_n = -C_n x_n \left[1 - 2x_n + \left(2 - \frac{1}{C_n} \right) x_n^2 + O(x_n^3) \right].}$$

In particular,

$$\beta_n \sim -(2^{n+2}\pi - 1) e^{-\pi 2^{n+1}}.$$

Moreover,

$$\boxed{\frac{\beta_{n+1}}{\beta_n^2} = -\frac{C_{n+1}}{C_n^2} \left[1 + 4x_n + \left(6 + \frac{2}{C_n} \right) x_n^2 + O(x_n^3) \right].}$$

Consequently,

$$\frac{\beta_{n+1}}{\beta_n^2} \sim -\frac{1}{\pi 2^{n+1}}, \quad \boxed{-\pi 2^{n+1} \frac{\beta_{n+1}}{\beta_n^2} \longrightarrow 1.}$$

Proof. By Lemma 3,

$$\beta_n = -C_n x_n \frac{1 + (2 - C_n^{-1})x_n^2 + O(x_n^4)}{1 + 2x_n + 4x_n^2 + O(x_n^4)}.$$

Since

$$\frac{1}{1 + 2x + 4x^2 + O(x^4)} = 1 - 2x + 8x^3 + O(x^4),$$

multiplication gives the first expansion.

Now $x_{n+1} = x_n^2$. Applying the first expansion at n and $n+1$ gives

$$\beta_{n+1} = -C_{n+1} x_n^2 [1 - 2x_n^2 + O(x_n^4)]$$

and

$$\beta_n^2 = C_n^2 x_n^2 \left[1 - 4x_n + \left(8 - \frac{2}{C_n} \right) x_n^2 + O(x_n^3) \right].$$

Taking the quotient and expanding the reciprocal yields the stated quotient formula. Finally,

$$\frac{C_{n+1}}{C_n^2} = \frac{2^{n+3}\pi - 1}{(2^{n+2}\pi - 1)^2} \sim \frac{1}{\pi 2^{n+1}},$$

which proves the scaled limit. $\square$

Remark 3. The theorem explains the formerly numerical observation that β_{n+1}/β_n^2 tends to zero with an approximately halving finite- n scale. Setting $A_n = 2^{n+2}\pi$,

$$\pi 2^{n+1} \frac{C_{n+1}}{C_n^2} = \frac{A_n(2A_n - 1)}{2(A_n - 1)^2} = 1 + \frac{3}{2A_n} + \frac{2}{A_n^2} + O(A_n^{-3}).$$

Thus

$$-\pi 2^{n+1} \frac{\beta_{n+1}}{\beta_n^2} = 1 + \frac{3}{2^{n+3}\pi} + O(4^{-n}) + O(x_n).$$

8 Relation to Static and Dynamic Cancellation Operators

The static Baker N-series cancellation operator targets even-power expansions in a scale parameter and uses fixed rational weights to cancel terms. The Gauss–Legendre error is doubly exponential with a rapidly varying prefactor, so no fixed rational weight yields exact cancellation across iterates. The dynamic fitted coefficient α_n plays the role of a local, data-driven cancellation weight, analogous in spirit to Aitken’s Δ^2 process which eliminates a single geometric error mode estimated from consecutive iterates.

9 Summary

1. Brent’s theta-function expansion yields the prefactor asymptotic $A_n \sim -2^{n+4}\pi^2$ for the lower Gauss–Legendre approximants.
2. Consequently $A_{n+1}/A_n \rightarrow 2$, and the fitted coefficients satisfy $\alpha_{n+1}/\alpha_n^2 \rightarrow 1/2$.
3. The limit $1/2$ arises from exact cancellation of the doubly-exponential factor and from the prefactor quotient; it is not a consequence of a fixed quadratic error map.
4. The numerator-channel coefficients satisfy

$$\beta_n \sim -(2^{n+2}\pi - 1)e^{-\pi 2^{n+1}}, \quad -\pi 2^{n+1} \frac{\beta_{n+1}}{\beta_n^2} \rightarrow 1.$$

Reproducibility. Numerical values were generated by reconstructing the Gauss–Legendre sequence at 500 decimal digits and computing the diagnostics $\varepsilon_n, \alpha_n, \alpha_{n+1}/\alpha_n^2, \varepsilon_{n+2}\varepsilon_n^2/\varepsilon_{n+1}^3$, and A_{n+1}/A_n with `mpmath`. A companion high-precision audit checks the β_n prefactor expansion and scaled quotient law. The reference π was `mp.pi`.

References

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